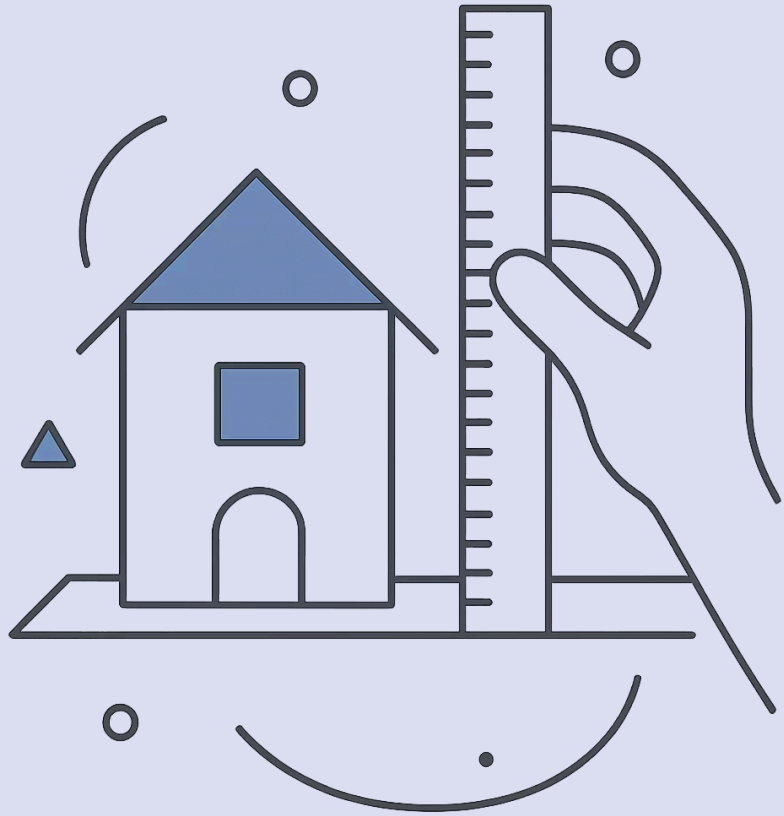




Regression Evaluation Metrics

How do we know if a regression model is good? We've learned models that predict numbers — Linear Regression, Decision Tree, Random Forest. Now the big question: how do we measure whether predictions are good or bad? That's where evaluation metrics come in.

Why Do We Need Evaluation Metrics?

Suppose a model predicts house prices. Some predictions are close, some are far — we need one number to measure overall performance.

House	Actual Price (₹ lakh)	Predicted Price (₹ lakh)
House 1	80	78
House 2	100	90
House 3	60	70

That single summary number is called an evaluation metric.

What Is an Evaluation Metric?

An evaluation metric tells how good or bad the model is by comparing actual value vs. predicted value.

- ① Actual price = ₹100 lakh
Predicted price = ₹90 lakh
Difference = ₹10 lakh error



What Is Error?

Error = Actual - Predicted

Example 1

Actual = 80

Predicted = 75

Error = +5

Example 2

Actual = 80

Predicted = 85

Error = -5

The sign shows direction — positive means under-prediction, negative means over-prediction.

Error Can Be Positive or Negative

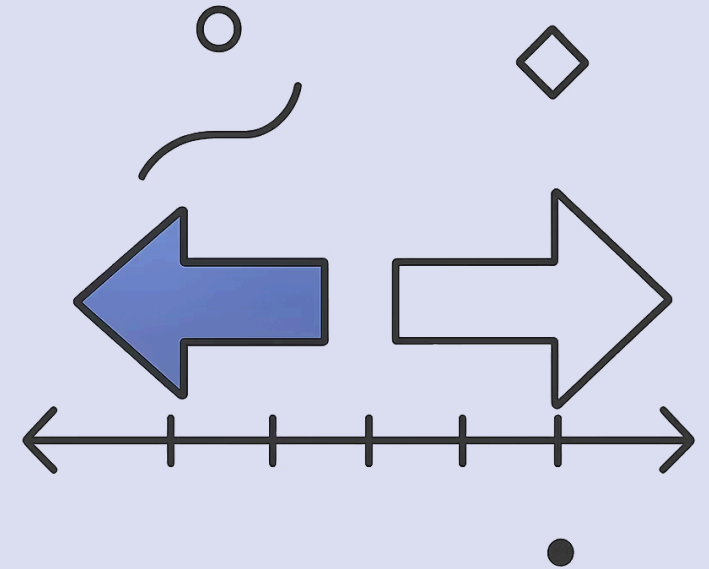
The negative sign only shows direction — the model predicted *more* than actual. During evaluation, we care about how big the mistake is, not just its direction. So we use:

Absolute Error

Ignore the sign — take the magnitude.

Squared Error

Square the error — always positive, punishes large mistakes.



Sample Data for Metrics

We'll use this example throughout. Prices are in ₹ lakhs.

House	Actual Price	Predicted Price
House 1	80	75
House 2	100	110
House 3	60	65
House 4	120	100

Error Calculation

House	Actual	Predicted	Error
House 1	80	75	+5
House 2	100	110	-10
House 3	60	65	-5
House 4	120	100	+20

Errors: 5, -10, -5, 20 — a mix of positive and negative values.

The Problem with Simple Average Error

If we directly average the errors:

⚠ $(5 + -10 + -5 + 20) \div 4 = 2.5 \text{ lakh}$

This says average error is only ₹2.5 lakh — but that's misleading. Positive and negative errors cancelled each other out.

Actual Mistakes Were:

5
lakh

10
lakh

5
lakh

20
lakh

We Need Better Metrics

Simple averaging hides the real mistake. Regression uses four proper evaluation metrics:

1

MAE

Mean Absolute Error

2

MSE

Mean Squared Error

3

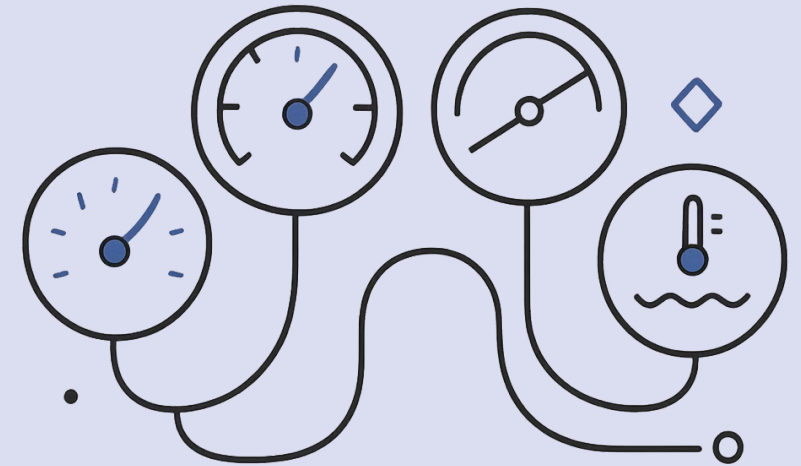
RMSE

Root Mean Squared Error

4

R^2 Score

Explained Variation



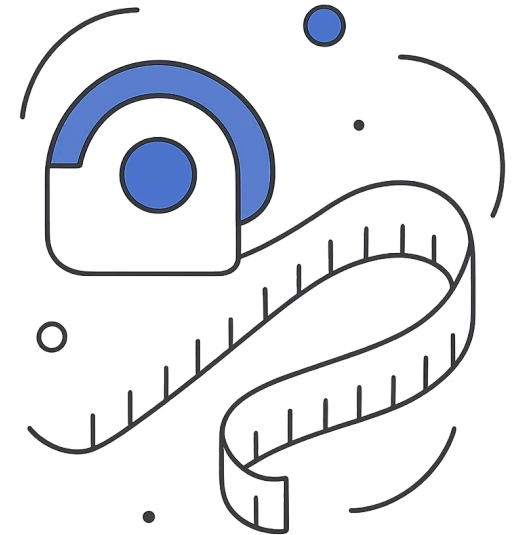
Metric 1 — MAE

MAE

Mean Absolute Error

Average absolute mistake made by the model. Absolute means: ignore the negative sign.

 Error = -10
Absolute Error = 10



MAE Calculation

House	Error	Absolute Error
House 1	+5	5
House 2	-10	10
House 3	-5	5
House 4	+20	20

✔ $MAE = (5 + 10 + 5 + 20) \div 4 = 40 \div 4 = 10$

How to Interpret MAE

MAE = 10

Average error = ₹10 lakh per prediction.

MAE = 3

Average error = ₹3 lakh per prediction.

✅ Lower MAE is better. MAE = 3 is better than MAE = 10.

Metric 2 — MSE

MSE

Mean Squared Error — average of squared errors.

Instead of absolute value, we square the error. Why?

- Makes negative errors positive
- Punishes large errors more strongly



MSE Calculation

Errors: 5, -10, -5, 20

Error	Squared Error
5	25
-10	100
-5	25
20	400

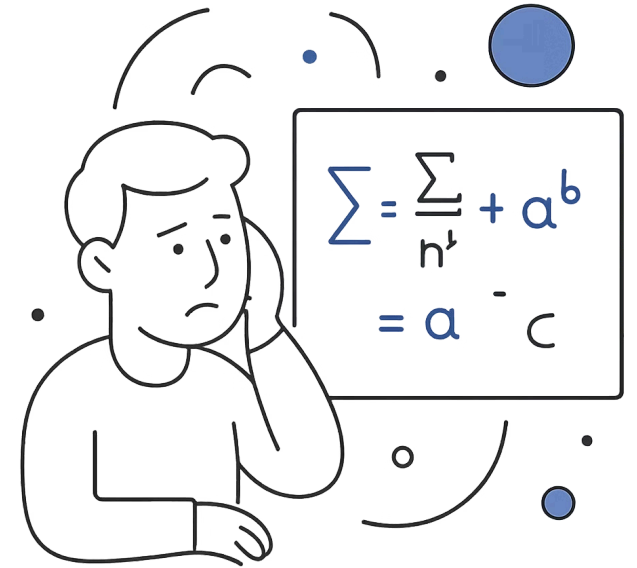
✔ $\text{MSE} = (25 + 100 + 25 + 400) \div 4 = 550 \div 4 = 137.5$

The Problem with MSE

MSE is useful mathematically, but harder to explain to beginners and business users.

⚠ If price is in lakhs, MSE is in lakh^2 — a squared unit that's hard to interpret.

That's why we often prefer RMSE — it brings the unit back to the original scale.

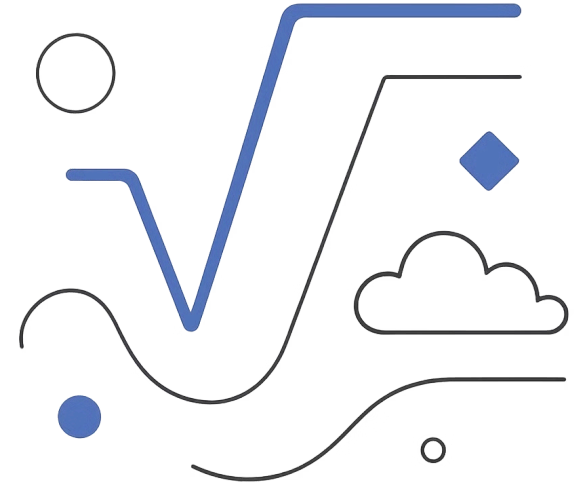


Metric 3 — RMSE

RMSE

Root Mean Squared Error — simply the square root of MSE.

 $RMSE = \sqrt{MSE}$
 $RMSE = \sqrt{137.5}$
 $RMSE \approx 11.73$



How to Interpret RMSE

RMSE is easier than MSE because it returns to the original unit.

Target Unit	RMSE Unit	Interpretation
Lakhs (₹)	Lakhs (₹)	RMSE = 11.73 → model error ≈ ₹11.73 lakh

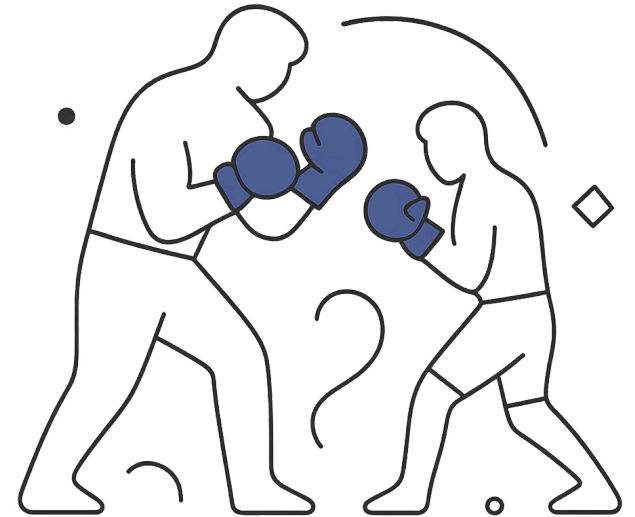
MAE vs. RMSE

MAE

- Average normal error
- Less affected by very large errors
- Easy to explain

RMSE

- Punishes large errors more
- Useful when big mistakes are serious
- Still in original unit



MAE vs. RMSE — A Concrete Example

Model	Errors	RMSE Impact
Model A	5, 5, 5, 5	Consistent — low RMSE
Model B	1, 1, 1, 17	One big mistake — RMSE penalizes heavily

Use RMSE when large errors are dangerous — e.g., medical dosage prediction, high-value property pricing, financial risk.

Metric 4 — R^2 Score

R^2 is different — it doesn't measure error. It measures how well the model explains variation in the target.

$$R^2 = 1.0$$

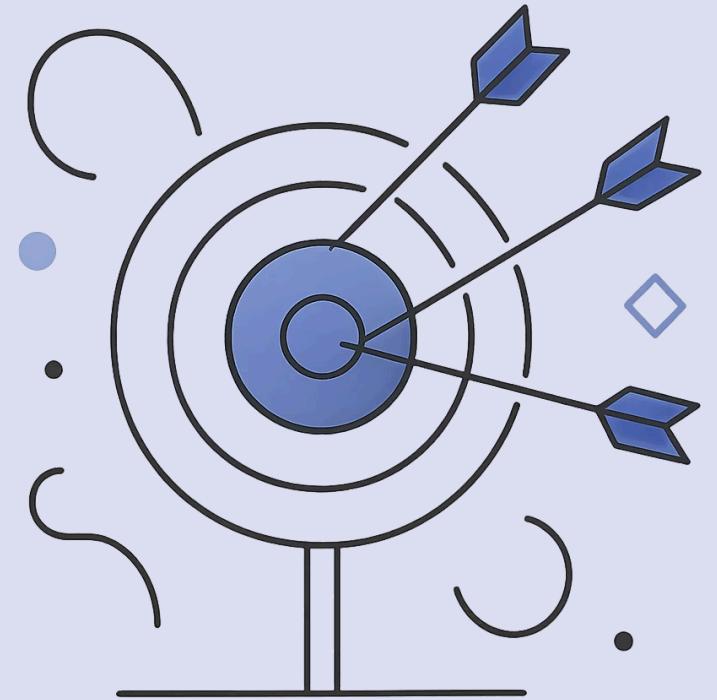
Perfect model

$$R^2 = 0.0$$

No better than predicting the average

$$R^2 < 0$$

Worse than predicting the average

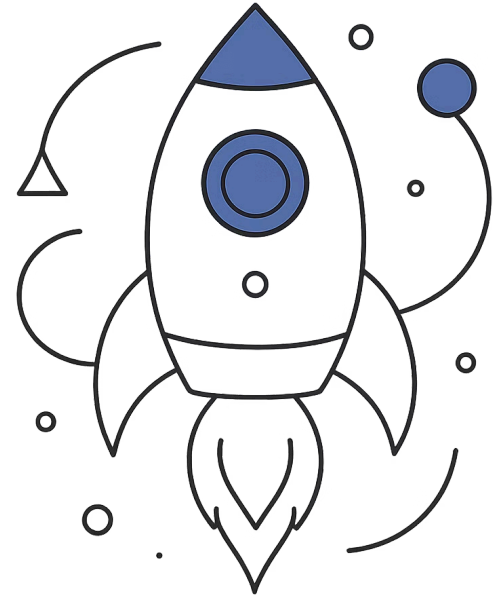


Simple R^2 Explanation

Suppose house prices are: 60, 80, 100, 120. A basic model always predicts the average = ₹90 lakh.

R^2 asks: *Is our model better than simply predicting the average every time?*

- ① Higher R^2 = model explains more variation than the average baseline.



R² Interpretation Table

R ² Score	Meaning
1.00	Perfect prediction
0.90	Very good
0.70	Good in many cases
0.50	Moderate
0.00	No better than average prediction
Negative	Worse than average prediction

 Higher R² is better — but never use R² alone. Always check error metrics too.

Metrics Direction Summary



MAE, MSE, RMSE

Lower is better. Smaller error = more accurate model.



R^2 Score

Higher is better. Closer to 1.0 = model explains more variation.



Simple memory: Error metrics → lower is better. R^2 score → higher is better.

Python Code for Metrics

```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

actual = [80, 100, 60, 120]
predicted = [75, 110, 65, 100]

mae = mean_absolute_error(actual, predicted)
mse = mean_squared_error(actual, predicted)
rmse = np.sqrt(mse)
r2 = r2_score(actual, predicted)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

Expected Output

MAE: 10.0

MSE: 137.5

RMSE: 11.73

R2 Score: (some value)

MAE = 10.0

Average absolute
mistake

RMSE = 11.73

Bigger mistakes
punished

MSE = 137.5

Squared mistake
average

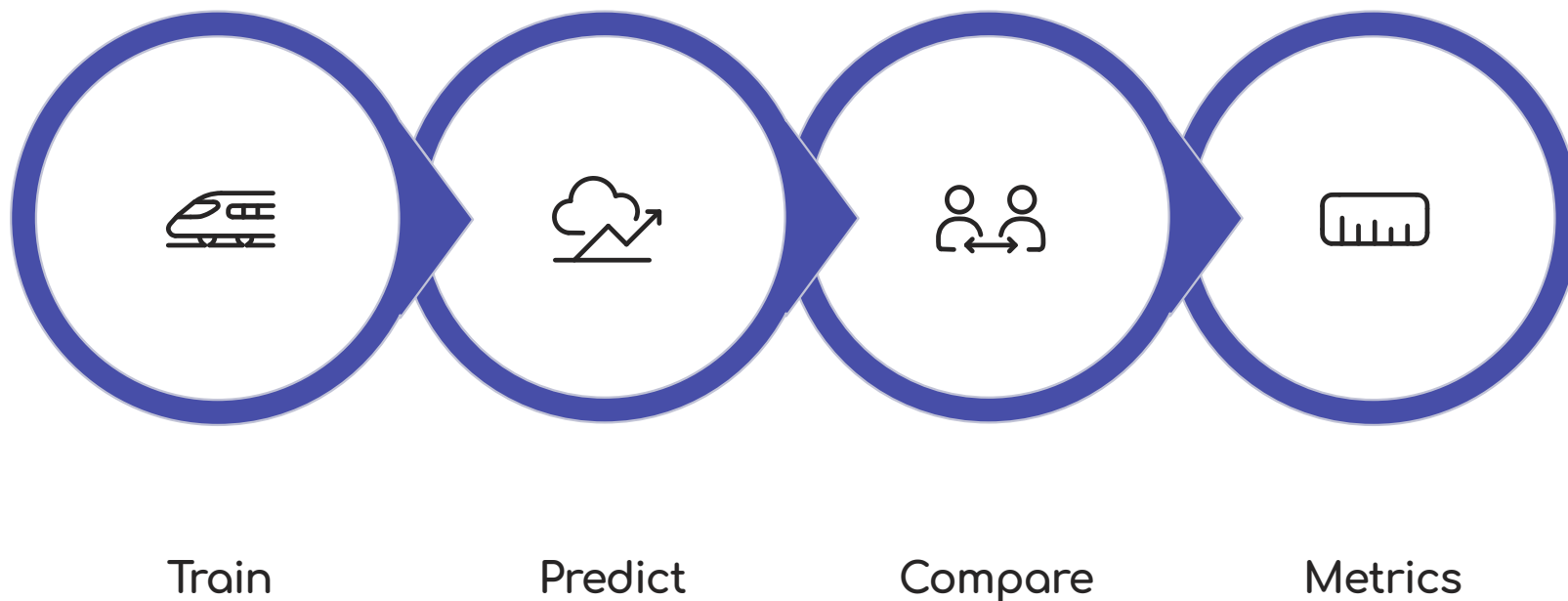
R^2

Model quality vs.
average prediction

Metrics in a Real Workflow

```
model.fit(X_train, y_train)  
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)  
mse = mean_squared_error(y_test, y_pred)  
rmse = np.sqrt(mse)  
r2 = r2_score(y_test, y_pred)
```



Best Order to Teach Beginners

01

MAE

Easiest — just the average mistake.

02

RMSE

Useful — punishes big mistakes, same unit as target.

03

R^2

Useful — shows how well model explains the target.

04

MSE

More mathematical — used internally and for optimization.



Business Explanation Example

Suppose your model gives: $MAE = 5$ lakh, $RMSE = 8$ lakh, $R^2 = 0.87$.
Here's how to explain it to a client:

"On average, our model is wrong by around ₹5 lakh. Because RMSE is ₹8 lakh, some larger mistakes exist. An R^2 of 0.87 means the model explains around 87% of price variation."

Final Summary

MAE

Average absolute mistake

MSE

Average squared mistake

RMSE

Square root of MSE

R^2

How well model explains target variation

✔ MAE, MSE, RMSE → lower is better | R^2 → higher is better

House Price Memory Hook

MAE = 10

Average error is ₹10 lakh.

RMSE = 11.73

Bigger mistakes are being punished.

$R^2 = 0.85$

Model explains 85% of variation.

NEXT TOPIC

Hands-on Regression Model Building Using Scikit-learn

